Requirement Elicitation

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall support student registration using Student ID, personal details, contact details, academic details, parent details, ID proof, hostel selection, and room allocation.	High	Students can register by providing the required registration and hostel details.	Enables students to access hostel-related services.
FR-002	Functional	The system shall allow students to view and edit their profile, including profile photo, room details, fee status, leave history, complaint history, and booking history.	Medium	Students can view and update their profile information and access the listed records.	Helps students manage their personal and hostel information.
FR-003	Functional	The system shall provide notifications and announcements, including notices, emergency alerts, fee reminders, maintenance alerts, and read/unread status.	Medium	Students can receive and view notifications and announcements with read/unread status.	Keeps students informed about important hostel updates.
FR-004	Functional	The system shall provide room management features including room details, capacity, bed availability, student allocation, room transfer, vacant rooms, and maintenance status.	High	Room information, availability, allocation, transfer, vacancy, and maintenance status can be managed and accessed.	Supports effective management of hostel rooms.
FR-005	Functional	The system shall allow students to book common facilities by viewing the facility list and availability, selecting a date/time slot, booking or cancelling, and viewing booking history and maintenance status.	Medium	Students can view available facilities, make or cancel bookings, and access booking history.	Helps manage the use of common hostel facilities.
FR-006	Functional	The system shall manage fee details, fee calculation, due dates, online payment, payment status, receipts, payment history, and fines/dues.	High	Students can view fee details, make online payments, check payment status, access receipts and payment history, and view fines or dues.	Provides organized management of hostel fee records and payments.
FR-007	Functional	The system shall provide mess management features including weekly menu, meal schedule, meal registration, meal attendance, mess feedback, mess complaints, mess holidays, and mess fee.	Medium	Students can access mess information and use the available mess-related features.	Helps students manage and access mess-related services.
FR-	Functional	The system shall allow	High	Students can submit	Provides a structured

008		students to apply for leave by specifying leave type, date/duration, reason, and parent contact, and allow leave status and history to be maintained.		leave applications and view their leave status and history.	process for managing student leave requests.
FR-009	Functional	The system shall support laundry management through laundry requests, item details, pickup, delivery, request status, charges, payment status, and laundry history.	Mediu m	Students can submit laundry requests and view request, payment, and laundry history details.	Helps manage hostel laundry services digitally.
FR-010	Functional	The system shall support lost and found services, including reporting lost or found items with item details, location, date/time, photo, search, and claim/return.	Low	Students can report, search, and claim or return lost and found items.	Provides a centralized way to manage lost and found items.
FR-011	Functional	The system shall allow students to submit complaints with category, description, attachment, priority, and status tracking.	High	Students can submit complaints with the required details and track their status.	Helps students report and track hostel-related issues.
NFR-001	Non-Functional	The system shall provide fast response time, quick page loading, multiple user support, fast search, and fast transaction processing.	High	The system supports responsive operation, search, transactions, and multiple users.	Ensures efficient system performance.
NFR-002	Non-Functional	The system shall provide a user-friendly interface with simple navigation, clear error messages, easy registration, easy booking, responsive design, and accessibility.	High	Users can access th main features throug a simple an understandable interface acros supported devices.	Provides a better user experience.
NFR-003	Non-Functional	The system shall provide user authentication, password protection, data encryption, secure payments, and unauthorized access prevention.	High	Users are authenticated, protected data is secured, and unauthorized access is prevented.	Protects user and system data.
NFR-004	Non-Functional	The system shall ensure system stability, error handling, data consistency, transaction reliability, and minimal downtime.	High	The system maintains consistent data and handles errors and transactions reliably.	Ensures dependable system operation.
NFR-005	Non-Functional	The system shall provide continuous system access, server availability, maintenance notifications, service continuity, and fault	High	Users can access the system with support for service continuity and maintenance notifications.	Improves system availability and continuity.

		tolerance.			
NFR-006	Non-Functional	The system shall support more students, increased user load, and database scalability.	Medium	The system can accommodate growth in students, users, and database requirements.	Allows the system to grow as usage increases.
NFR-007	Non-Functional	The system shall support easy bug fixing, code readability, easy updates, component reusability, logging and monitoring, and easy feature integration.	Medium	The system supports maintenance, monitoring, updates, bug fixes, and feature integration.	Makes the system easier to maintain and improve.
NFR-008	Non-Functional	The system shall maintain accurate student records and secure payment records.	High	Student and payment records are maintained accurately and securely.	Ensures data integrity.
NFR-009	Non-Functional	The system shall support web browsers, cross-device access, and operating system compatibility.	Medium	Users can access the system through supported web browsers on different devices and operating systems.	Ensures compatibility and portability.

Requirement Analysis

1. User Requirements

The Hostel Management System shall provide students with the ability to:

- Register using Student ID, personal details, contact details, academic details, parent details, ID proof, hostel selection, and room allocation.
- View and edit their profile and access room, fee, leave, complaint, and booking information.
- Receive notifications and announcements, including emergency alerts, fee reminders, and maintenance alerts.
- View room details, bed availability, room allocation, vacant rooms, and maintenance status.
- View common facilities, check availability, and book or cancel facilities.
- View fee details, make online payments, check payment status, and access receipts and payment history.
- Access mess information such as weekly menus, meal schedules, meal registration, feedback, and mess fees.
- Apply for leave by providing the required leave details and track leave status and history.
- Submit laundry requests and view pickup, delivery, charges, payment status, and laundry history.
- Report lost or found items, search for items, and claim or return them.
- Submit complaints with descriptions, attachments, priority, and track complaint status.

2. System Requirements

The Hostel Management System shall:

- Provide a web-based platform with a centralized database for managing hostel services and student information.
- Provide secure user authentication and password protection.
- Support student registration and profile management.
- Provide room, fee, mess, leave, facility booking, laundry, lost and found, complaint, and notification modules.
- Support online payments and maintain payment records.
- Provide fast response time, quick page loading, fast search, and fast transaction processing. • Support multiple users and increased user load.
- Provide a user-friendly, responsive, and accessible interface.
- Maintain data consistency, reliability, and integrity.
- Provide secure HTTPS communication for data transfer.
- Support desktop/laptop, tablet, and smartphone access.

3. Domain Requirements

The Hostel Management System shall comply with requirements specific to hostel management, including:

- The system shall follow institutional hostel rules and policies.
- Every student shall have a unique Student ID.
- Hostel and room information shall be maintained correctly.
- The system shall maintain consistency between related hostel records.
- Room capacity, bed availability, student allocation, room transfers, vacant rooms, and maintenance status shall be managed.
- Hostel fees, due dates, payments, receipts, payment history, and fines/dues shall be maintained.
- Leave applications shall include leave type, date/duration, reason, and parent contact.
- Mess-related information such as menus, meal schedules, attendance, feedback, holidays, and mess fees shall be managed.
- Hostel facilities shall have availability, booking slots, booking approval, and maintenance status.
- Hostel complaints shall support categorization, priority, attachments, and status tracking.